

Pathology foundation-model research-software demonstration

29 September 2026

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Research-software output: internally derived TCGA estimates

COAD - TCGA-AA-A01F - Prov-GigaPath

UNVALIDATED RESEARCH OUTPUT - REFERENCE RANK, NOT PROBABILITY - NO INDEPENDENT EXTERNAL VALIDATION - NOT FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION

Models applied	Continuous	Binary	Slides pooled
21	14	7	1

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Every displayed **Reference rank, not probability** describes relative position only.

For every cancer and representation, PathoFMPred displays only models that met the recorded representation-specific predictability rule. In the matched three-representation atlas, this means Q2 at least 0.20 for a continuous endpoint or AUROC at least 0.60 for a binary endpoint. TITAN tasks outside the common cohort must pass the supporting permutation/FDR screen. By default, the report also excludes continuous models developed with fewer than 100 labelled participants and binary models with fewer than 50 participants in either class. Explicit opt-in can add only these limited-sample threshold-crossing models. Tested-below-threshold reference objects never enter prediction reports.

Case context (not used as model input)

Shared selection features. Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins. Nodal status was not matched: example A was pN1, whereas the available slide summary for example B did not state a nodal category.

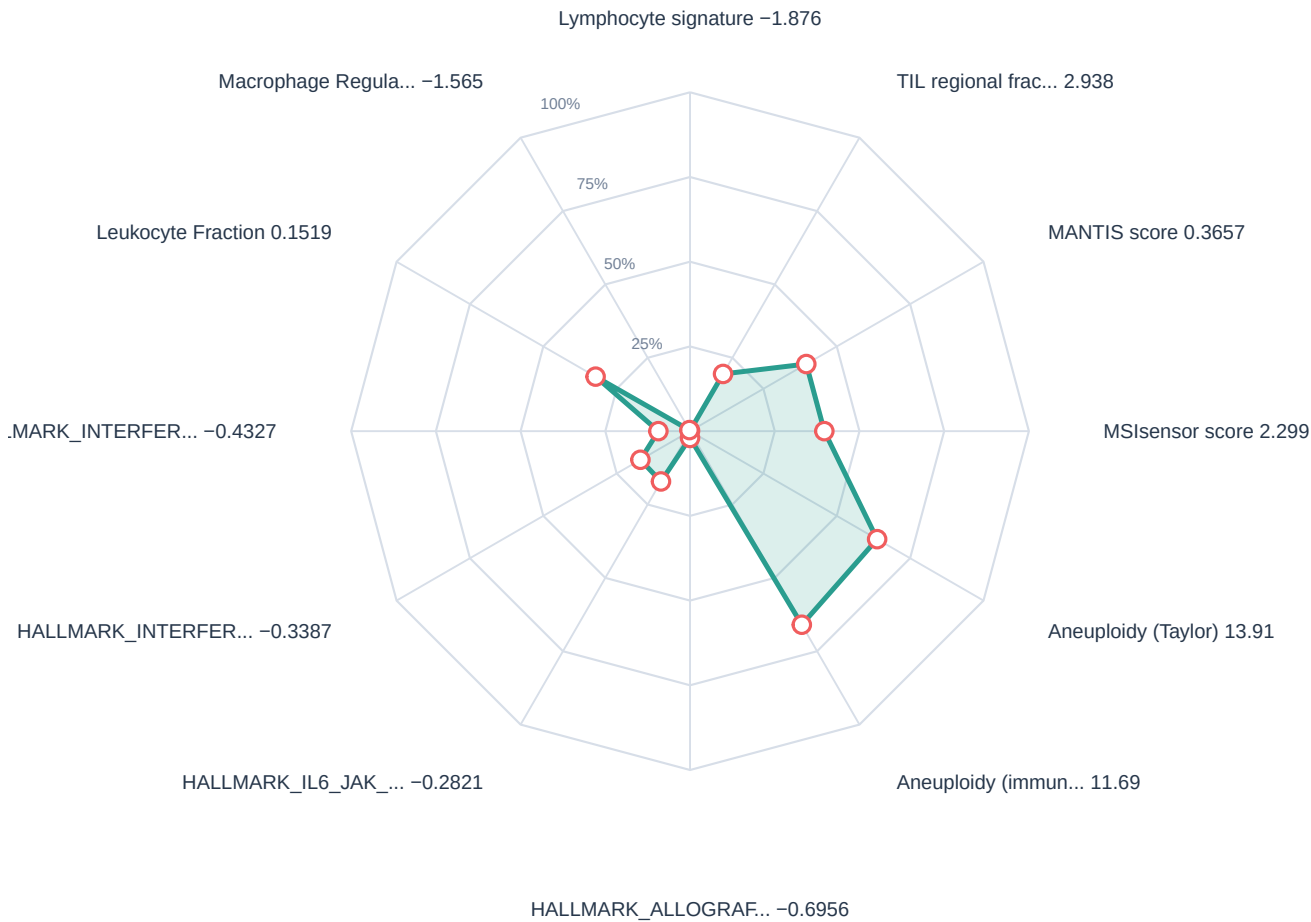
Pathology summary. The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Clinical-context source. Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

The pathology field is descriptive context only; it was not a predictor, training target, or validation outcome for PathoFMPred.

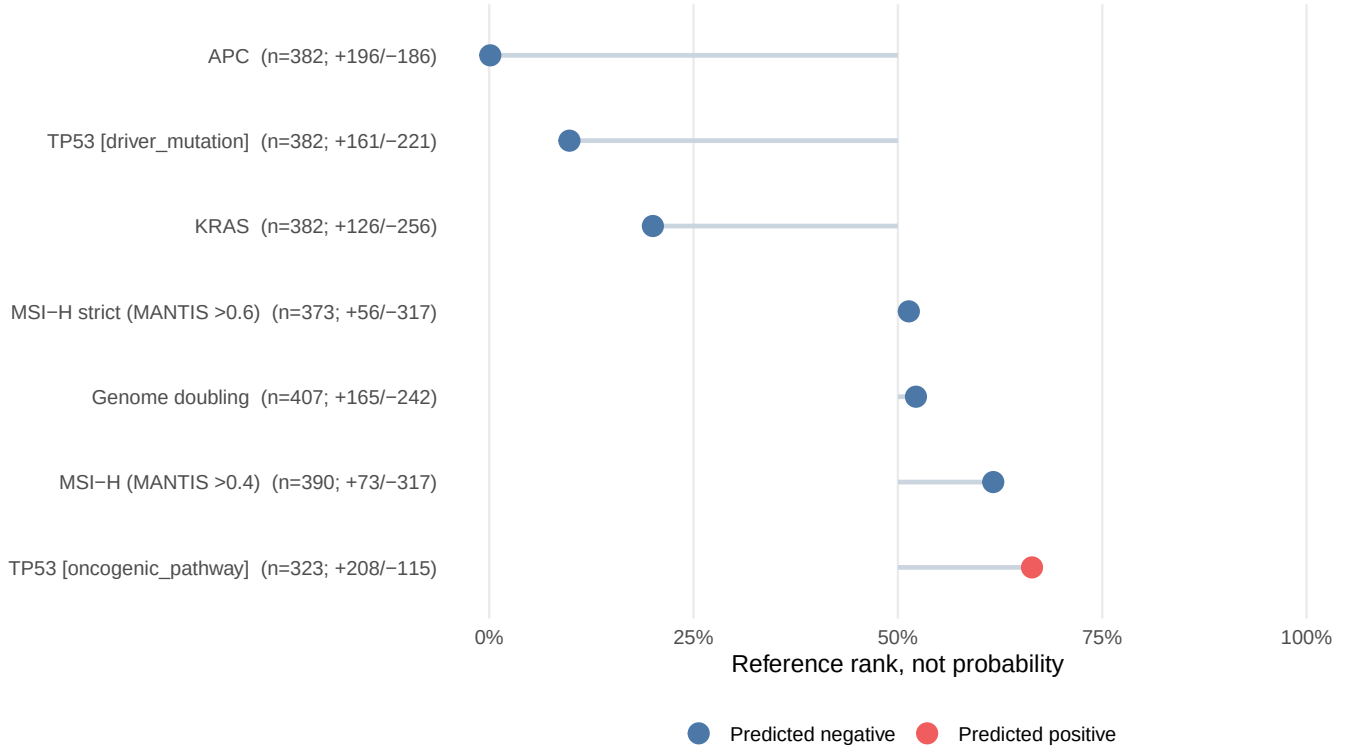
Continuous internally derived TCGA estimates

Only endpoints predictable for this cancer and representation are shown.



Binary research-model calls

Reference rank, not probability. Red points are predicted positive.



Continuous internally derived estimates

Family	Endpoint	Prediction	TCGA OOF rank (not probability)
thorsson	Lymphocyte Infiltration Signature Score	-1.876	0.1%
thorsson	Macrophage Regulation	-1.565	0.4%
rna_pathway_activity	HALLMARK_ALLOGRAFT_REJECTION	-0.6956	2.0%
rna_pathway_activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.4327	9.3%
rna_pathway_activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	-0.3387	16.9%
rna_pathway_activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	-0.2821	17.1%
thorsson	TIL Regional Fraction	2.938	19.5%
thorsson	Leukocyte Fraction	0.1519	32.2%
thorsson	Aneuploidy Score	11.69	66.0%
aneuploidy	Aneuploidy score	13.91	63.8%
microsatellite_instability	MANTIS score	0.3657	39.6%
microsatellite_instability	MSIsensor score	2.299	39.6%
thorsson	SNV Neoantigens	4.249	49.3%
aneuploidy	Deleted arm count	6.413	50.5%

Internal development evidence. The primary screen and five-repeat estimates use different nested-CV partitions and need not be identical.

Endpoint	Primary screen Q2	Five-repeat Q2	Five-repeat RMSE	Five-repeat Spearman
Aneuploidy score	0.294	NA	NA	NA
Deleted arm count	0.205	NA	NA	NA
MANTIS score	0.300	NA	NA	NA
MSIsensor score	0.254	NA	NA	NA
HALLMARK_ALLOGRAFT_REJECTION	0.318	NA	NA	NA
HALLMARK_IL6_JAK_STAT3_SIGNALING	0.244	NA	NA	NA
HALLMARK_INTERFERON_ALPHA_RESPONSE	0.231	NA	NA	NA
HALLMARK_INTERFERON_GAMMA_RESPONSE	0.284	NA	NA	NA
Aneuploidy Score	0.296	NA	NA	NA
Leukocyte Fraction	0.274	NA	NA	NA
Lymphocyte Infiltration Signature Score	0.311	NA	NA	NA
Macrophage Regulation	0.291	NA	NA	NA
SNV Neoantigens	0.271	NA	NA	NA
TIL Regional Fraction	0.507	NA	NA	NA

Binary research-model calls

Calls and development denominators.

Endpoint	Class	LDA score	Threshold	Rank, not probability	Training n (+/-)
Genome doubling	0	-0.515	-0.5093	52.2%	407 (+165/-242)
APC	0	-9.439	0.9281	0.1%	382 (+196/-186)
KRAS	0	-3.535	-1.586	20.0%	382 (+126/-256)
TP53 [driver_mutation]	0	-4.388	-1.325	9.8%	382 (+161/-221)
MSI-H (MANTIS >0.4)	0	-3.124	-1.882	61.7%	390 (+73/-317)
MSI-H strict (MANTIS >0.6)	0	-6.191	-4.648	51.3%	373 (+56/-317)
TP53 [oncogenic_pathway]	1	2.33	-0.001568	66.4%	323 (+208/-115)

Internal development evidence. Primary AUROC and PR-AUC use nested held-out scores. The no-skill PR-AUC reference equals prevalence. Operating metrics and predictive values use training-derived thresholds.

Endpoint	AUROC	PR-AUC	Prevalence	No-skill PR-AUC	Threshold BA
Genome doubling	0.765	0.701	0.405	0.405	0.698
APC	0.842	0.796	0.513	0.513	0.768
KRAS	0.740	0.526	0.330	0.330	0.682
TP53 [driver_mutation]	0.787	0.695	0.421	0.421	0.729
MSI-H (MANTIS >0.4)	0.822	0.618	0.187	0.187	0.764
MSI-H strict (MANTIS >0.6)	0.857	0.620	0.150	0.150	0.840
TP53 [oncogenic_pathway]	0.724	0.809	0.644	0.644	0.657

Cohort-specific predictive values.

Endpoint	PPV	NPV	Sample-size stratum
Genome doubling	0.623	0.749	Larger-sample stratum
APC	0.759	0.851	Larger-sample stratum
KRAS	0.580	0.785	Larger-sample stratum
TP53 [driver_mutation]	0.710	0.824	Larger-sample stratum
MSI-H (MANTIS >0.4)	0.662	0.894	Larger-sample stratum
MSI-H strict (MANTIS >0.6)	0.710	0.929	Larger-sample stratum
TP53 [oncogenic_pathway]	0.755	0.638	Larger-sample stratum

Binary development reliability

Fold class counts and component selection.

Endpoint	Minimum outer test (+/-)	Minimum inner training (+/-)	Components: median (range)
Genome doubling	+33 / -48	+105 / -155	8.0 (2-16)
APC	+39 / -37	+124 / -118	2.0 (1-5)
KRAS	+25 / -51	+80 / -163	4.0 (2-20)
TP53 [driver_mutation]	+32 / -44	+102 / -140	2.0 (2-7)
MSI-H (MANTIS >0.4)	+14 / -63	+47 / -202	6.0 (4-13)
MSI-H strict (MANTIS >0.6)	+11 / -63	+36 / -202	8.0 (5-16)
TP53 [oncogenic_pathway]	+41 / -23	+132 / -73	3.0 (1-7)

Repeat-to-repeat held-out prediction stability.

Endpoint	Score Spearman rho	Class agreement
Genome doubling	0.760	0.829
APC	0.943	0.942
KRAS	0.732	0.837
TP53 [driver_mutation]	0.929	0.923
MSI-H (MANTIS >0.4)	0.801	0.919
MSI-H strict (MANTIS >0.6)	0.759	0.945
TP53 [oncogenic_pathway]	0.862	0.900

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 4.3%.
- Foundation-model input schema: model-specific named features validated against the fitted artifact.
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- TCGA tissue-source-site-code metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-code count, threshold-retention status, fold-size range and inner-code-separation audit.
- Binary reliability metadata: held-out average precision, cohort-specific PPV/NPV, outer and inner class-count minima, selected-component distribution and repeat-to-repeat prediction stability.
- Default model selection: binary models require at least 50 TCGA participants in each class; smaller-class models are available only by explicit opt-in.
- Scope: internal validation grouped by TCGA tissue-source-site code is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

Endpoint interpretation and target sources

For continuous endpoints, the TCGA OOF prediction percentile (not probability) locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, every Reference rank, not probability value is the relative position of an uncalibrated LDA score. It is not confidence or a clinical-risk estimate. PR-AUC is non-interpolated average precision for positive class 1. PPV and NPV are held-out estimates at the observed endpoint prevalence in this TCGA analysis and will change when prevalence changes. Rows marked exploratory/limited evidence are excluded from default inference and require explicit opt-in.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **TP53 [driver_mutation].** Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score

Model-target source: MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.

- **MSI-H (MANTIS >0.4).** Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6).** Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.
- **TP53 [oncogenic_pathway].** Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score.** Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **Deleted arm count.** Number of chromosome arms classified as deleted; larger values indicate a greater arm-level deletion burden. *Analysed scale:* source endpoint units *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score.** Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale:* source endpoint units *Model-target source:* Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score.** Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale:* source endpoint units *Model-target source:* Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **HALLMARK_ALLOGRAFT_REJECTION.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_IL6_JAK_STAT3_SIGNALING.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_INTERFERON_ALPHA_RESPONSE.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_INTERFERON_GAMMA_RESPONSE.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **Aneuploidy Score.** Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Leukocyte Fraction.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **Lymphocyte Infiltration Signature Score.** Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Macrophage Regulation.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.

- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. PathoFMPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* log1p-transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.